

# Best-fit lines, outliers and conclusions

Working scientifically  
Both papers · Combined and Separate  
10 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                             | ANSWER                                                                               | ✓                        |
|----|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------|
| 1  | What is a line of best fit?                                          | One smooth line showing the overall trend of the data.                               | <input type="checkbox"/> |
| 2  | Describe how a line of best fit should be drawn.                     | As one thin line with points balanced above and below it.                            | <input type="checkbox"/> |
| 3  | State what must <b>not</b> be done when drawing a line of best fit.  | Joining the points dot-to-dot.                                                       | <input type="checkbox"/> |
| 4  | What is an outlier?                                                  | A result that does not fit the pattern of the others.                                | <input type="checkbox"/> |
| 5  | Describe what you should do when you find an outlier.                | Check the recording and plotting, then repeat the measurement. Never just delete it. | <input type="checkbox"/> |
| 6  | What is a trend?                                                     | The overall relationship shown by the data.                                          | <input type="checkbox"/> |
| 7  | State what a good conclusion must include.                           | The relationship between the variables, supported by values with units.              | <input type="checkbox"/> |
| 8  | Describe the graph of two quantities that are directly proportional. | A straight line through the origin.                                                  | <input type="checkbox"/> |
| 9  | State the three parts of a good evaluation point.                    | The problem, its effect on the data, and an improvement that fixes it.               | <input type="checkbox"/> |
| 10 | What is peer review?                                                 | When other scientists check the method, the evidence and the claims.                 | <input type="checkbox"/> |